

Paper Directions 20 — my read (Aug 13, 2026)

Source: `transcripts/20260813_Paper_Directions_20_otter_ai_transcript.docx` (~5 min follow-up, same day as PD19).
Context: Charles briefed Alex well on the LG three-step pipeline (ASSIGN → SCORE → SELECT). PD19 opened **probabilistic SELECT + MLE**; PD20 locks **temperature convention** and **reorders priorities** (inverted-U survival before MLE).
PD19 digest: `transcripts/PD19_notes.md`
Related PD19 artifacts (same day):

Artifact	Path
Transcript (PD19)	<code>transcripts/20260813_Paper_Directions_19_otter_ai_transcript.docx</code>
Formula synopsis	<code>3-Master_Plan/Alex_stuff/MLE_prob.docx</code>
Whiteboard	<code>3-Master_Plan/Alex_stuff/PD19_board.jpeg</code>
Briefing doc	<code>3-Master_Plan/re_entry/HEROs_and_PASSes/Alex_LG_three_step_briefing.md</code>

Headline

Replace deterministic top-K SELECT with a **Gibbs (softmax) rule at temperature t** , with t in the denominator (statistical-physics convention). **First job**: prove the **inverted-U still emerges** at some t when everything else is held at the PD17/LG settings that worked. **MLE for $\rho^*, \lambda^*, \gamma^*$ is step two** — not blocked, but not first.

Arc — PD19 → PD20

PD19 (morning meeting + board)

- SCORE unchanged**: $S_i = \hat{A}_i - \lambda L_{C,g(i)}$ (team smooth congestion).
- SELECT change**: instead of top-K by S_i , define $P(Y_i = 1)$ from the selection score.
- MLE ambition**: fit $\rho^*, \lambda^*, \gamma^*$ (and likely t) to real draft outcomes.
- Synopsis doc** wrote $P(Y = 1|i) \propto \exp(t(\hat{A}_i - \lambda L_C)) / \sum_i \exp(\dots) - t$ in the numerator (non-physics convention).
- Board (bottom left)** looked like t in the denominator — PD20 confirms that is the lock.

PD20 (follow-up call)

- Alex's **first-order worry**: softening the hard top-K cut may **smooth away the breakpoint** that produces the inverted-U ("lose a phase transition").
- Hope**: in the **right t regime**, Gibbs SELECT $\approx$ hard cut, but t adds a **knob** on U shape.
- Lock**: t in denominator — Gibbs / statistical-physics temperature.
- Immediate work**: sim + **log-spaced t sweep**; reaffirm U; then MLE / full parameter story.

Locked formula — Gibbs SELECT

Selection score (unchanged — SCORE step):

$$S_i = \hat{A}_i - \lambda L_{C,g(i)}$$

Team congestion (unchanged):

$$L_{C,j} = \frac{1}{|j|} \sum_{k \in j} \sigma(\gamma(\hat{A}_k - \theta)), \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

Selection probability (SELECT step — PD20 lock):

$$P(Y_i = 1) = \frac{\exp(S_i/t)}{\sum_{k=1}^N \exp(S_k/t)}$$

Alex framing: Gibbs probabilities — S_i plays the role of a **Hamiltonian** (objective / energy-like score); t is **temperature**. Same family as stat-phys / transfer-matrix intuition (Laplace transform link came up in passing).

Temperature behavior (t in denominator)

Regime	Effect on SELECT	Link to old rule
$t \rightarrow 0^+$ (cold)	Weight concentrates on highest $S_i \rightarrow \approx$ deterministic top-ranked	Recovers hard cut
t moderate	Soft ranking; tunable sharpness	Target operating band for inverted-U
t large (hot)	$\approx$ uniform over pool	Noise dominates; U may flatten

Sweep t logarithmically — Alex: orders of magnitude in the exponent matter; don’t only linspace small values.

What stays the same (BINDING)

Per `BINDING_Selection_is_its_own_step.md`:

Separation	PD20 status
ASSIGN vs SCORE vs SELECT	Still three steps
Score $\neq$ select	S_i built in SCORE; only SELECT rule changes
λ lives in SCORE	Unchanged
ρ lives in ASSIGN	Unchanged (LG Grandchild)
Hero	Layer A outcome — not the scoring equation

MLE does not merge layers — it estimates parameters **inside** the generative story once the SELECT rule is validated.

Alex priority order (explicit)

1. **Build a SELECT rule** from Gibbs probabilities that is **equivalent to top-K** in the cold- t limit.
2. **Simulate** — hold ASSIGN + SCORE at the **PD17/LG settings that produced the inverted-U** (empirical caps, ρ , λ band, etc.).
3. **Sweep t** (log scale) — plot selection rate vs LOO pool quality (Hero axis).
4. **Success criterion:** inverted-U **re-emerges** at some t ; hard-cut behavior at small t .
5. **Then** — MLE for ρ^* , λ^* , γ^* (and t); “rest of” PD19 agenda.

Not first: full MLE pipeline before U is reaffirmed.

Charles test plan (from call — Alex said “sure”)

Panel	Fix	Sweep
A	$\lambda \approx 1$ (congestion in score)	t log-spaced
B	λ at breakpoint (where λ sweep first moved SELECT vs talent-only)	t log-spaced

Question both panels answer: Does t kill the inverted-U that λ created, or is there a t window where the hump survives?

Implementation notes (for code — not yet built)

Bridging Gibbs to K draft slots

PD20 did **not** re-litigate the “softmax sums to 1” vs “K winners per season” issue. Practical bridge for **sim readouts** (consistent with “equivalent to top-K at cold t ”):

- Compute weights $w_i \propto \exp(S_i/t)$ with **log-sum-exp** for stability.
- **Draw K players without replacement** proportional to w_i (Plackett-Luce / rule `"A"` skeleton in `tier1_pool_assignment.choose_selected`).
- Bin realized `Y_selected` vs `poolq_loo` — same Hero comparison as `grandchild_lambda_select_sweep.py`.

Flag for Alex (one line email): “Sim uses K draws without replacement from Gibbs weights — OK?”

MLE (PD19 — parked until step 5)

When ready, likelihood on player-seasons:

$$\mathcal{L} = \prod_i P(Y_i = 1 \mid S_i(t, \lambda, \gamma, \dots))^{Y_i} (1 - P(Y_i = 1 \mid \dots))^{1-Y_i}$$

ρ in MLE: only enters if rosters are **re-simulated** (LG assign). On **fixed empirical rosters**, ρ does not move L_C — match ρ via H_{sort} separately (Alex loved H_{sort} for the paper).

ASSIGN recap (unchanged — for PD20 reader)

LG weight (unnormalized): $\tilde{w}_{ij} = R_j \exp(-\rho|\hat{A}_i - \hat{\mu}_j|)$.

Seat probability: $P(j|i) = \tilde{w}_{ij} / \sum_{j'} \tilde{w}_{ij'}$.

Each player **draws** a team (weighted random seat). Final $\hat{T}_j = \mu_j$.

Open questions (carry forward)

#	Question	Owner
1	K draws without replacement vs independent Bernoulli scaled to K ?	Confirm with Alex (one line)
2	Is expected selection curve (analytical P) enough, or only simulated draws?	Charles / first plots
3	θ still $F^{-1}(1 - K/N)$ per season when sweeping t ?	Default yes unless MLE pass
4	Exact log-t grid (e.g. 10^{-3} to 10^3)?	Empirical after first run

Code touchpoints (when Charles says go)

Piece	File / pattern
SELECT rule "D" (Gibbs + K draws)	<code>sports/tier1_pool_assignment.py</code> → <code>choose_selected</code>
t sweep diagnostic	New script mirroring <code>grandchild_lambda_select_sweep.py</code>
Baseline λ sweep (already done)	<code>sports/scripts/grandchild_lambda_select_sweep.py</code>
LG ASSIGN	<code>sports/541_grandchild_homophily_assign.py</code>
Briefing / notation	<code>3-Master_Plan/re_entry/HEROs_and_PASSes/Alex_LG_three_step_briefing.md</code>

One-liners for the paper / defense

Gibbs SELECT: "Advancement probability follows a Gibbs rule on the selection score S_i , with temperature t ; the deterministic top-K limit is the cold- t regime."

Why PD20 before MLE: "We first verify that the phenomenological inverted-U is not an artifact of the hard cut alone — it must survive a soft Gibbs SELECT in a finite t window before we estimate parameters by likelihood."

H_{sort} **(PD19 carryover):** Realized sorting diagnostic for ASSIGN — Alex wants it in the paper; separate from SELECT temperature.

Next doc (optional)

Mirror to re_entry when stable: `3-Master_Plan/re_entry/10_PD20_Alex_meeting_takeaways.md` (COMPASS-style, like `09_PD18`).